



Formation mechanisms of TiB₂ tracks on Ti6Al4V alloy during selective laser melting of ceramic-metal multi-material

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ABSTRACT

Multi-material selective laser melting (SLM) is a new development direction in additive manufacturing technologies to realize the incorporation of different materials within a single component. In this work, processing parameters, cross-sectional features and surface morphologies of TiB₂ tracks on SLM-processed Ti6Al4V alloy layer were investigated. A processing window was determined by examining the type of molten pools and continuity of scanned tracks. Formation mechanism of microstructures at the cross-sections of TiB₂ tracks was also discovered. At a high scan speed above 700 mm/s, the tracks all exhibited an unstable surface consisting of excessive shrinkage, spatters and cracks. Consequently, based on the experimental analysis and numerical simulations, a range of laser energy density from 0.67 J/mm to 1.50 J/mm at laser powers of 400–450 W was found to be a feasible scope of processing parameters to fabricate stable tracks of TiB₂ on Ti6Al4V alloy by SLM.

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1. Introduction

As a layer-by-layer additive manufacturing technique, selective laser melting (SLM) is capable of fabricating complex components by melting metallic, ceramic, and composite powders [1,2]. In metallic materials, the molten titanium is highly sensitive and reactive to oxygen, carbon and hydrogen, which make it hard for conventional processes. SLM-processed Ti6Al4V fabricated in inert gas environment can obtain high specific strength, excellent corrosion resistance, and superb biocompatibility [3]. Therefore, it has been widely used in aerospace, automatic and biomedical industries [4]. Titanium diboride (TiB₂), as a typical hard ceramic with high melting point, good wear resistance and great thermal stability [5,6], has been mainly used for extreme applications including electrode materials, cutting tools and engines. Noticeably, SLM is also a candidate to fabricate composites consisting of metal matrix and ceramic reinforcements. For example, the SLM-processed TiB/Ti6Al4V composites exhibit superior nanohardness and wear resistance due to the in-situ synthesized TiB reinforcement [7].

Except for SLM of the above single materials, a critical requirement in laser additive manufacturing is to increase the functionalities of the high-performance components for different applications. Hence, multi-material SLM has gradually attracted much attention.

For instance, Wei et al. have successfully fabricated 316 L/In718 and 316 L/Cu10Sn samples with a good metallurgical bonding at the material interfaces via selective laser melting [8]. Their research indicated that the interface design with special materials was vital to improve elemental diffusion and material bonding. A large amount of Fe and Cu element diffusion was observed at the SLM-processed interface of stainless steel and copper alloy, indicating a good metallurgical bonding in metal-metal interface [9]. These related studies have clearly revealed the feasibility of combining different metallic materials with good metallurgical bonding. Nevertheless, compared with the metal-metal multi-material, SLM of ceramic-metal multi-material has more challenges owing to the extremely different bonding strength and thermal expansion coefficients of them [10]. These distinctly physical properties could easily induce interfacial defects including porosity and cracks. Accordingly, we selected the TiB₂ and Ti6Al4V for SLM of ceramic-metal multi-material. It is not only due to their wide applications, but also they have almost similar density and coefficient of thermal expansion [5,11,12]. This special material design is helpful to enhance the interfacial bonding by reducing thermal stress at interface.

Owing to the track-by-track and layer-by-layer process of SLM, tracks formation is the basic element in SLM process. In this situation, the resultant density and the performance of complex components depend largely on the stability of each scanned track [13–15]. Specifically, the scanned tracks of titanium alloy [14], copper alloy [15] and aluminum alloy [16] all could be characterized as

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continuous, discontinuous and unstable under different processing parameters. Zhou et al. [14] determined the best processing parameters for 3D Ti-13Nb-13Zr samples based on the scanned tracks and deposited layer formation features. The optimally fabricated Ti-13Nb-13Zr component with high relative density had a higher hardness and tensile strength than that prepared by traditional techniques. That was because the cross-sectional feature and surface morphology of each scanned track also had a strong effect on the performance of the metal samples fabricated by SLM [17]. Thijs and Kempen [18,19] also investigated the effects of processing parameters on the formation of AlSi10Mg tracks. They found that the good surface quality of scanned track and even fully dense components with good mechanical properties could be fabricated using optimal processing parameters. Moreover, the common defects such as fragment and porosity were more likely to form in tracks [13]. When the laser energy density was low, the molten pool could not spread completely, and the energy could not penetrate the powder layer totally. Hence, lots of residual pores, many unmelted particles and balling were easy to form on the surface of the specimens, resulting in a rough surface [20].

According to the above analysis, it can be concluded that scanned track is significant to ceramic-metal multi-material additive manufacturing. This paper was to investigate the effects of processing parameters on geometrical characteristics of the molten pool and surface morphology of TiB₂ scanned tracks on SLM-processed Ti6Al4V alloy layer. The defects on the cross-sections and the surfaces of scanned tracks were analyzed through optical microscope and scan electron microscope techniques. A 3D multilayer bed model of TiB₂ and Ti6Al4V was established by ANSYS software to study the forming process of scanned track. Finite element analysis

(FEA) simulations of temperature distribution and cooling rate during SLM process were given.

2. Materials and methods

2.1. Powder preparation

Particle size and shape examined by scan electron microscopy (SEM) were shown in Fig. 1a and b. The starting materials were spherical Ti6Al4V powder with a 99.7% purity and irregular-shaped TiB₂ powder with a 98.9% purity. Clearly, these two powders had a distinctly different distribution curves. Ti6Al4V powder was characterized by a Gaussian like particle size distribution with a slight asymmetry. The Ti6Al4V powder with a homogeneous particle size distribution ($D_{10} = 16.58 \mu\text{m}$, $D_{50} = 28.64 \mu\text{m}$, $D_{90} = 52.93 \mu\text{m}$) was favorable to smooth spreading during SLM process. The fine TiB₂ powder with the mean sizes of about 3 μm contained 10% particles less than 1 μm , which could realize high powder bed density.

2.2. SLM processing

Selective laser melting experiments were performed using the multi-material SLM equipment, which was developed at Nanjing University of Aeronautics and Astronautics (NUAA). This equipment was composed of an IPG YLR-500 Ytterbium fiber laser with a $1070 \pm 10 \text{ nm}$ wavelength and a spot size of 70 μm , a computer system for process control, and an inert argon gas circulatory protection system. Fig. 2a showed the SLM process of different powders by two recoaters. Ti6Al4V blocks were prepared layer by layer on a Ti6Al4V substrate. An “island” scan strategy with small thermal stress was applied for

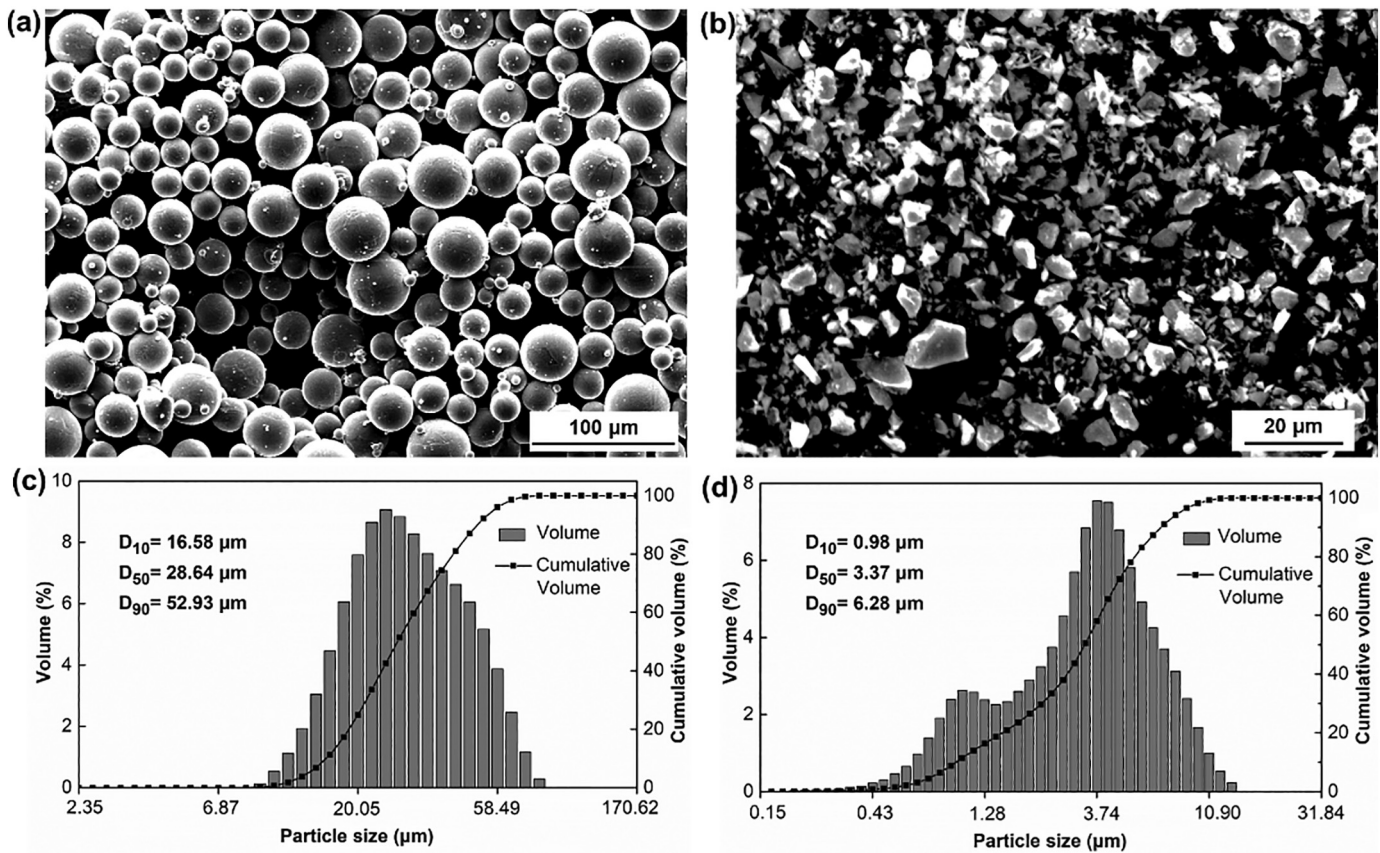


Fig. 1. Morphology and particle size distribution of powder: (a), (c) Ti6Al4V powder, and (b), (d) TiB₂ powder. Grey columns show particle size distribution at different scopes, black circles show cumulative distribution.

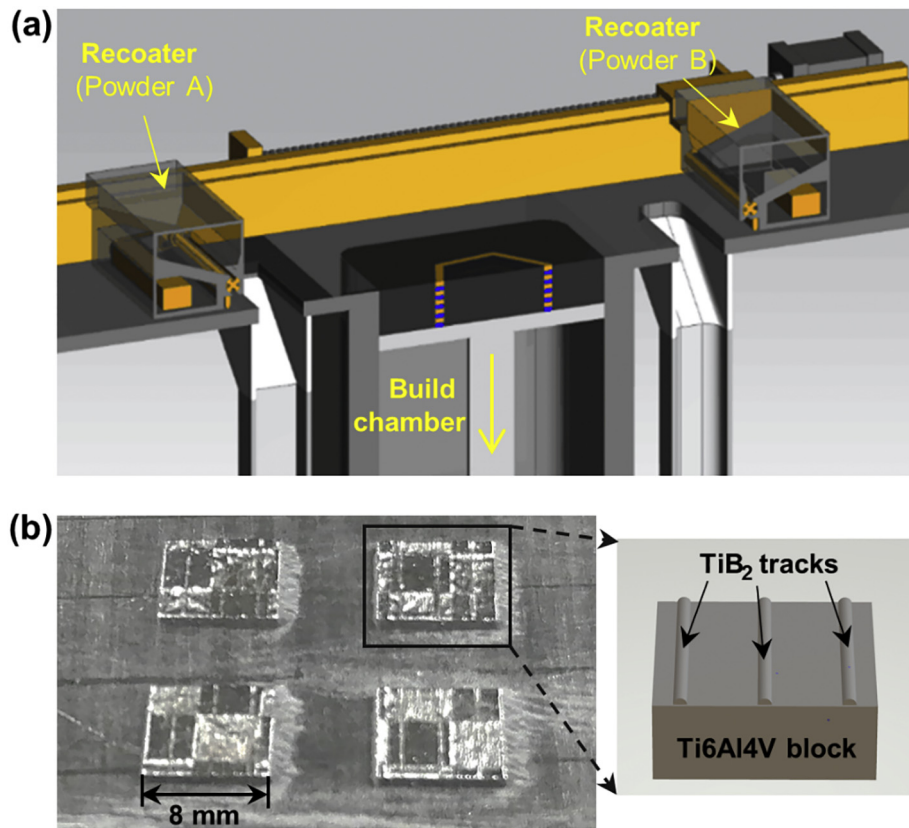


Fig. 2. (a) Schematic diagram of multi-material SLM equipment; (b) TiB_2 scanned tracks on SLM-processed Ti6Al4V alloy layer.

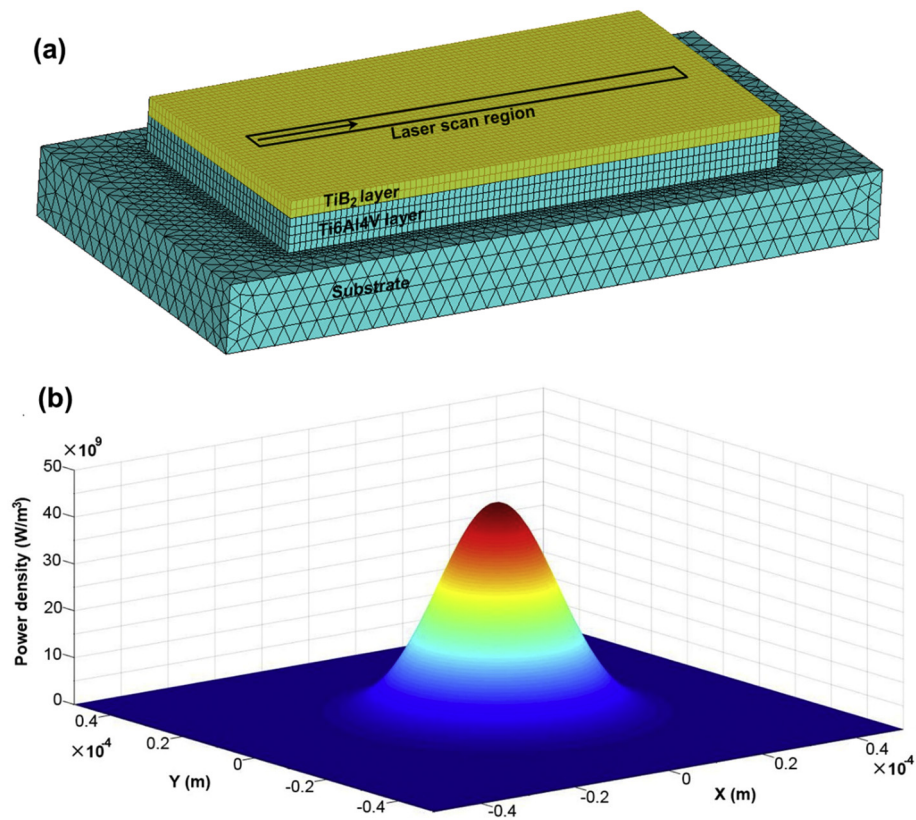


Fig. 3. (a) 3D finite element model; (b) Gaussian laser energy distribution.

Table 1

Thermal physical parameters of Ti6Al4V and TiB₂. k_s means conductivity (W/m·K) and c means specific heat (J/kg·K).

Ti6Al4V [25,26]			TiB ₂ [27]		
T (°C)	k_s (W/m·°C)	c (J/kg·°C)	T (°C)	k_s (W/m·°C)	c (J/kg·°C)
20	7.20	580	20	96.0	617
200	9.44	605	200	90.0	800
500	13.30	700	500	81.0	1073
870	18.20	930	1000	78.1	1160
1000	19.70	936	2000	77.3	1296
1655	28.27	1126	3000	77.1	1400

SLM. The laser power (P), scan speed (v), layer thickness, and hatch spacing of Ti6Al4V layer were 250 W, 800 mm/s, 50 μ m and 50 μ m, respectively. For the TiB₂ layer, the layer thickness was 50 μ m. laser scan speed varied from 100 mm/s to 900 mm/s with an interval of 200 mm/s, and the laser power varied from 250 W to 500 W with an interval of 50 W. During SLM process, the substrate was preheated to 240 °C in order to reduce the thermal stress.

2.3. Characterization

The particle size distributions of Ti6Al4V and TiB₂ powders were measured by a BT-9300S laser diffraction sensor (Bettersize Instruments Ltd., China). The produced samples were cut, ground and polished by colloidal silica suspension according to the standard metallographic procedure. The images of samples at cross-sections were characterized by PMG3 optimal microscope (Olympus Corporation, Japan) after the planes were etched in solution composing of HF, HNO₃ and distilled water with a volume ratio of 1:3:50 for 30 s. The top surfaces of the TiB₂ tracks were imaged by a FEI Quanta 200 scan electron microscopy.

2.4. Numerical simulations

Numerical simulations were performed to obtain the temperature distribution and cooling rate of the SLM-processed multi-material by ANSYS. A three-dimensional finite element model and Gaussian laser source during SLM process are shown in Fig. 3. The model consisted of TiB₂ powder layer, Ti6Al4V powder layer, and Ti6Al4V substrate with

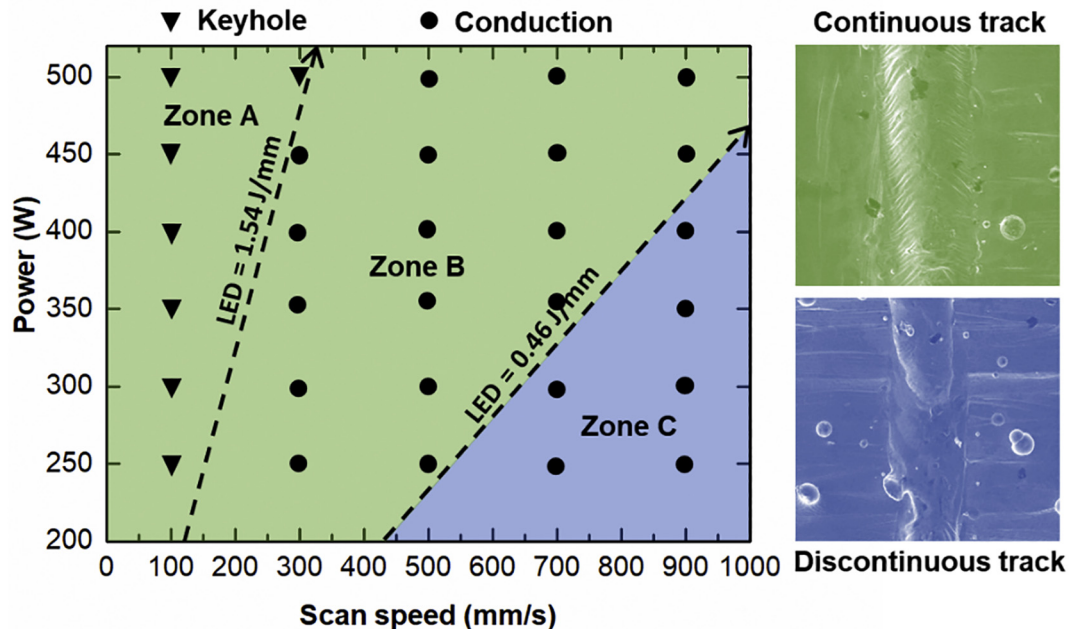


Fig. 4. Processing window for TiB₂ scanned tracks on SLM-processed Ti6Al4V layer in P - v space.

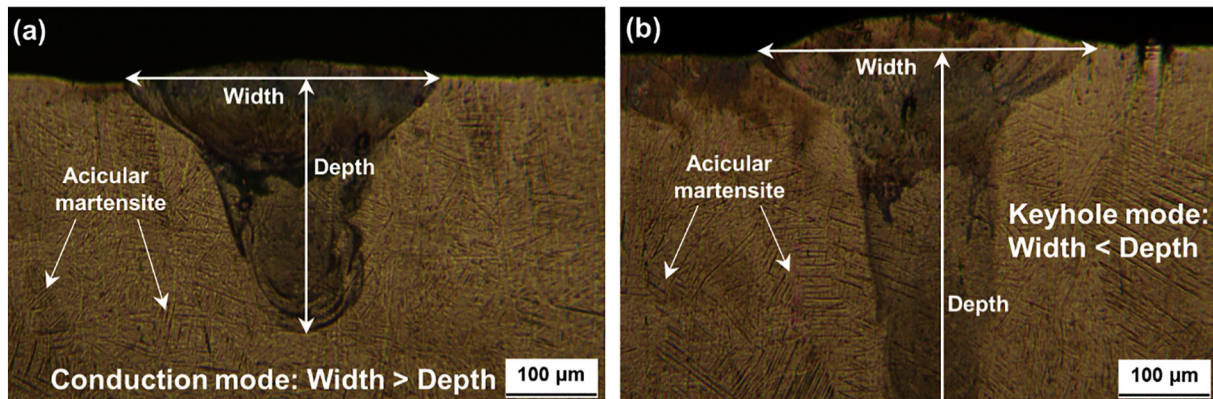


Fig. 5. OM images showing the cross-sections of scan tracks under different processing parameters: (a) Conduction mode at P of 450 W, v of 300 mm/s; (b) Keyhole mode at P of 500 W, v of 300 mm/s.

the dimensions of 1.50 mm × 0.75 mm × 0.05 mm, 1.50 mm × 0.75 mm × 0.10 mm, and 2.00 mm × 1.00 mm × 0.20 mm (Fig. 3a). Considering the simulation precision and computing efficiency, a comparatively coarse tetrahedron mesh was adopted in Ti6Al4V substrate. The powder bed was meshed using the Solid70 hexahedron elements with the dimensions of 17.5 μm × 17.5 μm × 25.0 μm. In this model, simulation of the multilayers used the element birth and death, which has been adopted to model the 3D temperature field in multi-layers during selective laser melting [21]. The moving laser heat source with a spot size of 70 μm put on the powder bed was considered as a Gaussian heat source distribution, which is shown in Fig. 3b. It can be presented as [22]:

$$q = \frac{2AP}{\pi R^2} \exp\left(-\frac{2r^2}{R^2}\right) \quad (1)$$

where A represents the laser energy absorptivity of the powder, specifically 0.77 for Ti6Al4V and 0.80 for TiB₂ [23]. P represents the laser power. R is the radius of Gaussian laser beam, and r is the radial distance from a point on the powder bed surface to the center spot of the laser beam. In this simulation, different laser powers of 250 W and 450 W and laser scan speeds of 300 mm/s and 900 mm/s were set for SLM process. The thermo-physical parameters of Ti6Al4V and TiB₂ are provided in Table 1. Details of other basic setups and governing equations can be found in our previous research [24].

3. Results and discussion

3.1. Processing window of scanned tracks

The processing window for TiB₂ scanned tracks on SLM-processed Ti6Al4V alloy layer is presented in Fig. 4. It is clear that the range of laser power is smaller than the laser scan speed. The linear energy density (LED) is adopted to evaluate the combined effect of scan speed and laser power on the stability of TiB₂ tracks, which is introduced by the following equation:

$$LED = P/v \quad (2)$$

where P is laser power, and v is laser scan speed. Discontinuous scanned tracks were found in the blue region with a low LED of 0.46 J/mm. They might strongly affect the forming defects of 3D components. For example, interlayer pores were mainly attributed to the discontinuous scanned tracks at low LED, which decreased the density and mechanical properties of SLM-processed samples [13]. With v decreasing or P increasing, the continuous tracks were observed in the green region with high LED. This phenomenon could be owing to the different laser-powder interaction radius caused by higher LED. It is worth noting that a transformation in the molten pool from conduction mode to key-hole mode occurred at the LED of 1.54 J/mm, namely, the depth of the molten pool exceeded the width of the molten pool with the increasing of LED (Fig. 5). Hence, as shown in Fig. 4, the P - v space could be divided

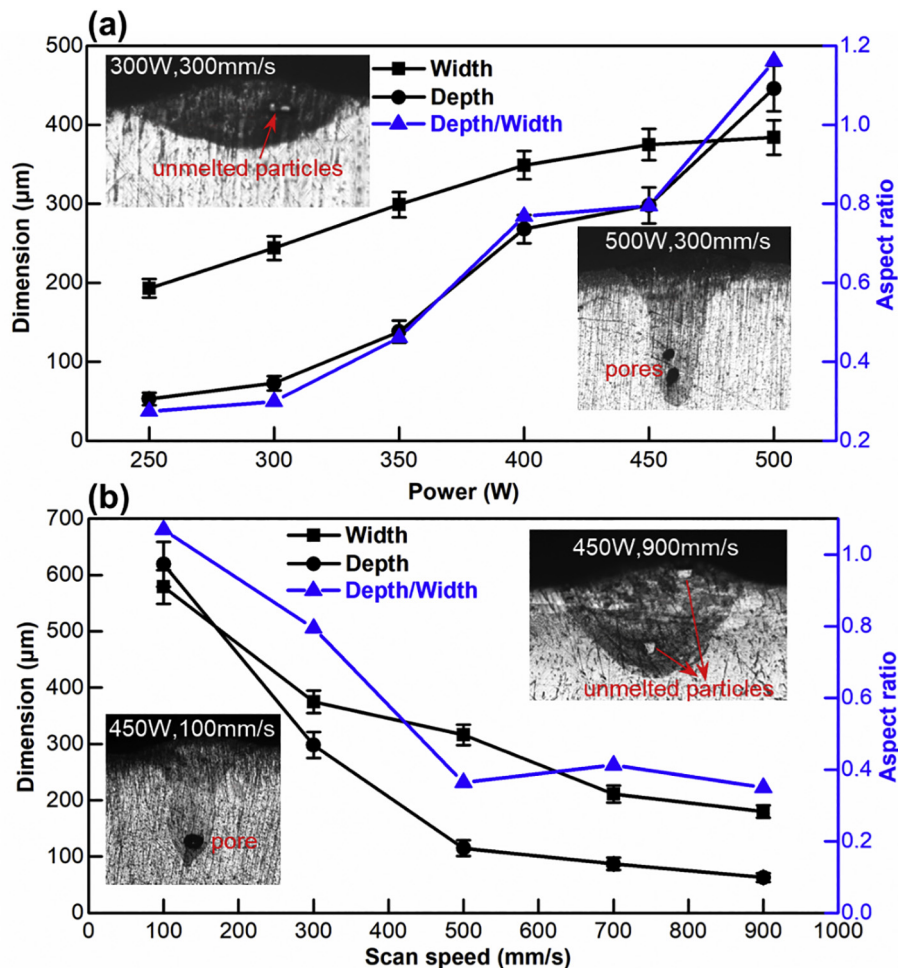


Fig. 6. (a) Effects of laser power on the width, depth, and aspect ratio (depth/width) under v of 300 mm/s; (b) Effects of scan speed on the width, depth, and aspect ratio (depth/width) under P of 450 W. The insets are the OM images of cross-sections at different processing parameters.

into three different categories: continuous scanned tracks with keyhole mode (Zone A), continuous scanned tracks with conduction mode (Zone B), and discontinuous scanned tracks with conduction mode (Zone C). Among them, zone A and zone C both are unstable regions. Zone B with LED of 0.46–1.54 J/mm are continuous TiB₂ tracks with conduction mode on SLM-processed Ti6Al4V alloy layer.

The molten pool mode of TiB₂ scanned tracks on SLM-processed Ti6Al4V was determined by the processing parameters. Two distinctive modes of conduction and keyhole modes can be observed because of the different LED applied to the SLM process. Fig. 5a shows the conduction mode at a relatively low LED. It is obvious that the width of the TiB₂ molten pool is larger than the depth. On the contrary, Fig. 5b suggests a keyhole mode of heat transfer when the laser power increases, which is characterized by a deep-narrow shape of molten pool. Actually, the shape of the TiB₂ molten pool depends on the laser-powder interaction. When the energy transfer efficiency was larger than the material's absorptivity, keyhole mode was formed in the molten pool and led to a creation of deep and narrow molten pool [28]. Keyhole mode melting induced the possibility of keyhole instabilities, which resulted in porosity formation. Furthermore, the acicular martensites were formed near the conduction mode and keyhole mode molten pool because cooling rate in SLM of Ti6Al4V is higher than 410 °C/s [29]. This is in agreement with Thijs's finding that the shape of martensites under different SLM processing parameters tends to be acicular [30].

3.2. Cross-sectional features of scanned tracks

The width, depth, and aspect ratio of the TiB₂ molten pool have a close relationship with laser power and scan speed. The width and depth of scanned tracks and the zone of powder consolidation were analyzed (Fig. 6). The width and depth increased along with the power increasing or the scan speed decreasing, and change of the depth was much larger than the width. Hence, comparing with the width of the

molten pool, penetration depth was more sensitive to the processing parameters. Specially, Fig. 6a depicts the width varied from 190 μm to 380 μm and the depth varied from 50 μm to 450 μm with the *P* increasing. The depth was larger than the width of scanned tracks as the *P* reached 500 W, because high LED (>1.54 J/mm) at high laser power made the molten pool transform from conduction mode to keyhole mode. In other words, elliptical section at *P* of 300 W and *v* of 300 mm/s changed into deep-narrow section at *P* of 500 W and *v* of 300 mm/s. In contrast to the variation of *P*, Fig. 6b depicts the width varied from 180 μm to 580 μm and the depth varied from 63 μm to 620 μm with the *v* decreasing. The high LED at relatively low scan speed resulted in a transition in the molten pool shape from elliptical to deep-narrow. In general, the increase of LED, whatever the laser power increasing or scan speed decreasing, leads to the increasing of the width and depth of scanned tracks as well as the growth of aspect ratios. From the insets of the Fig. 6, it was clear that the unmelted particles within conduction mode molten pool at relatively low LED and black pores within keyhole mode molten pool at relatively high LED both were detrimental to the final properties of SLM-processed components. Especially, the keyhole mode molten pool in laser additive manufacturing is usually associated with excessive porosity because of the entrapped gas in very deep layers. Therefore, in a view of cross-sections, a relatively moderate LED is significant to the fabrication of TiB₂ scanned tracks on Ti6Al4V layers, even for SLM of TiB₂/Ti6Al4V multi-material.

In order to reveal the element distribution and microstructure of the scanned track at cross-section, a TiB₂ track with *P* of 400 W and *v* of 500 mm/s was selected as a typical sample. As shown in Fig. 7a, some unmelted particles can be observed at the upper part of the TiB₂ molten pool, and the B element are concentrate on the inner of TiB₂ molten pool (Fig. 7b). The larger particle size (about 30 μm) of unmelted TiB₂ than that of starting powder was ascribed to the agglomeration of the TiB₂ powder during laser radiation. The occurrence of small amount of Al and V composition in the pool was induced by the strong Marangoni

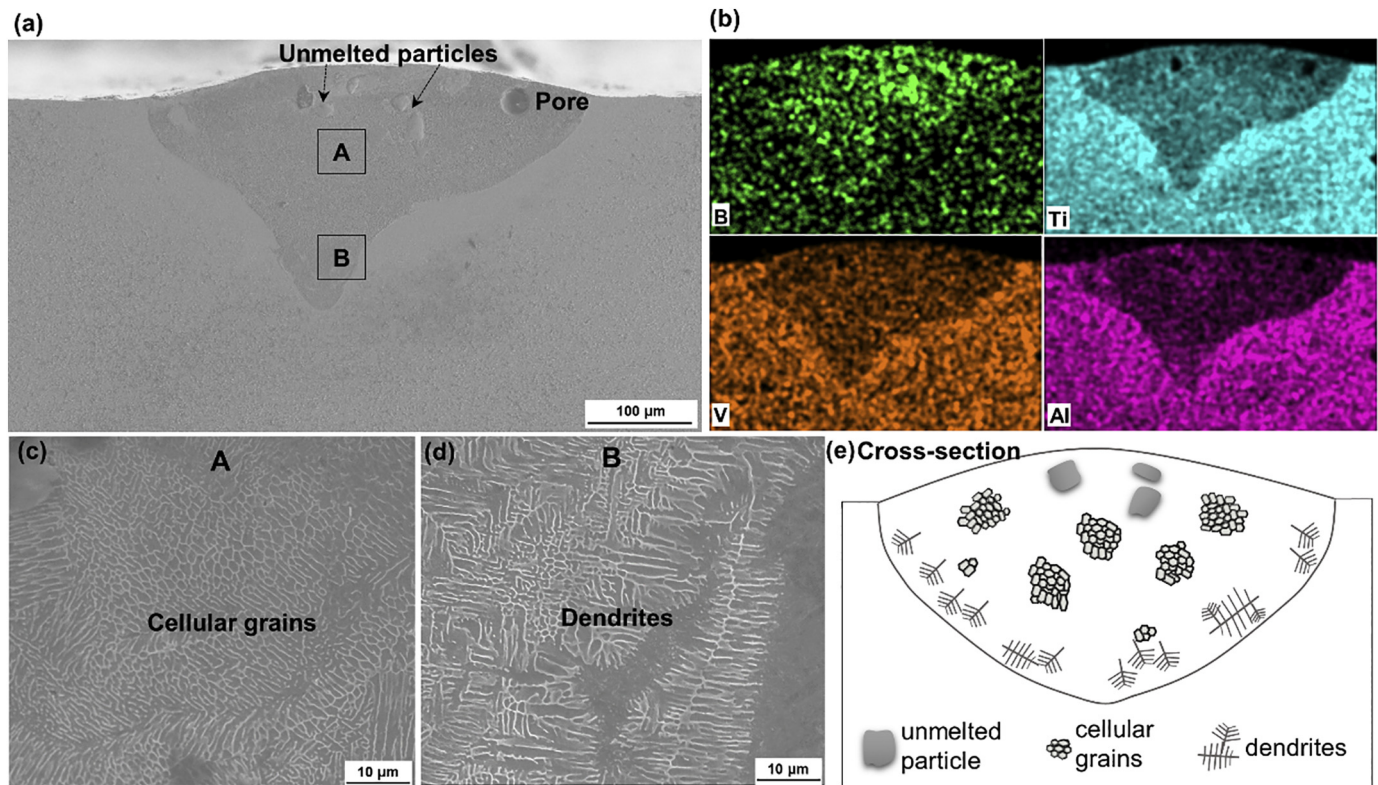


Fig. 7. (a) SEM images showing the cross-section of TiB₂ track under *P* of 400 W, *v* of 500 mm/s; (b) EDS mapping of the cross-section; (c) and (d) showing the microstructures of the regions A and B in (a); (e) showing the schematic of cross-sectional microstructures of TiB₂ track on Ti6Al4V layer by SLM.

flow in the molten pool, which was melted by the high-energy laser beam. Fig. 7c and d clearly depict the fine grains in the center and the boundary of the track section. Comparing with the cellular grains in the center of the molten pool, dendrites with elongated grain shape was attributed to the large temperature gradient around the boundary. For this reason, it can be observed that the fine cellular grains grew in the molten pool uniformly and the columnar grains with epitaxial growth from substrate were parallel to the temperature gradient. A similar microstructure distribution has been observed in Al-Cu-Mg alloy and stainless steel during SLM [31,32], they discovered that the type of microstructure depends on the cooling rate of materials. Thus, the boundary of the TiB_2 molten pool has higher temperature gradient and cooling rate resulting in elongation of the grains compared to the inner of the molten pool, which may be helpful for the good bonding between TiB_2 and Ti6Al4V. It can be concluded that the formation mechanisms of microstructure in the cross-section of TiB_2 on Ti6Al4V layer are as shown in Fig. 7e. The unmelted particles are distributed at the upper of TiB_2 molten pool, the cellular grains are occupied in the center of the pool, and the dendrites are spread over the boundary.

3.3. Surface morphologies of scanned tracks

Fig. 8 shows that the laser power has a significant influence on the surface stability of TiB_2 scanned tracks on SLM-processed Ti6Al4V layer. When the laser irradiated on the powder bed at a constant scan speed, the materials absorbed the laser energy and started to melt. For a relatively low laser power, the obtained surface roughness of samples was obviously larger than that at high laser power. For example, the

unmelted particles and folded surfaces were found in the surface of scanned tracks at P of 250 W–350 W. When the laser power was 250 W, the unmelted particles caused by insufficient energy input resulted in the cracks. As shown in Fig. 8a, these transverse cracks appeared due to the larger residual stress perpendicular to the scan direction than that along the scan direction [33]. When the laser power increased to 350 W, the balling phenomenon appeared on the surface, which might give rise to the poor interline bonding. In addition, owing to the layer-by-layer process, balling phenomenon could be a severe impediment to a uniform deposition of the next TiB_2 powder layer on the previous melted layer and tend to cause porosity and delamination. As laser power reached to 400 W, it was clear that the top surface was completely dense, showing no apparent unmelted particles or cracks. However, it was interesting to find that some spherical particles generated on the junction of two islands by analyzing the two different directions of yellow arrows (Fig. 8c). This phenomenon was ascribed to the thermal stress concentration at the boundary of islands, and obviously, occurrence of an uneven surface further led to the poor TiB_2 powder spreading on Ti6Al4V layer. For a relatively high power of 450 W, an even smoother surface can be observed in Fig. 8d because of the prolonged liquid lifetime. As a typical ceramic with high melting point and low ductility, it makes sense that TiB_2 powder requires relatively high laser power to increase the temperature of molten pool to melting point during the SLM process.

Fig. 9 depicts surface morphologies of samples at different scan speeds. When the scan speed was around 300–500 mm/s, a smooth surface morphology could be obtained at the relatively low scan speeds (Fig. 9a). From the other point, the liquid viscosity of TiB_2 scanned tracks

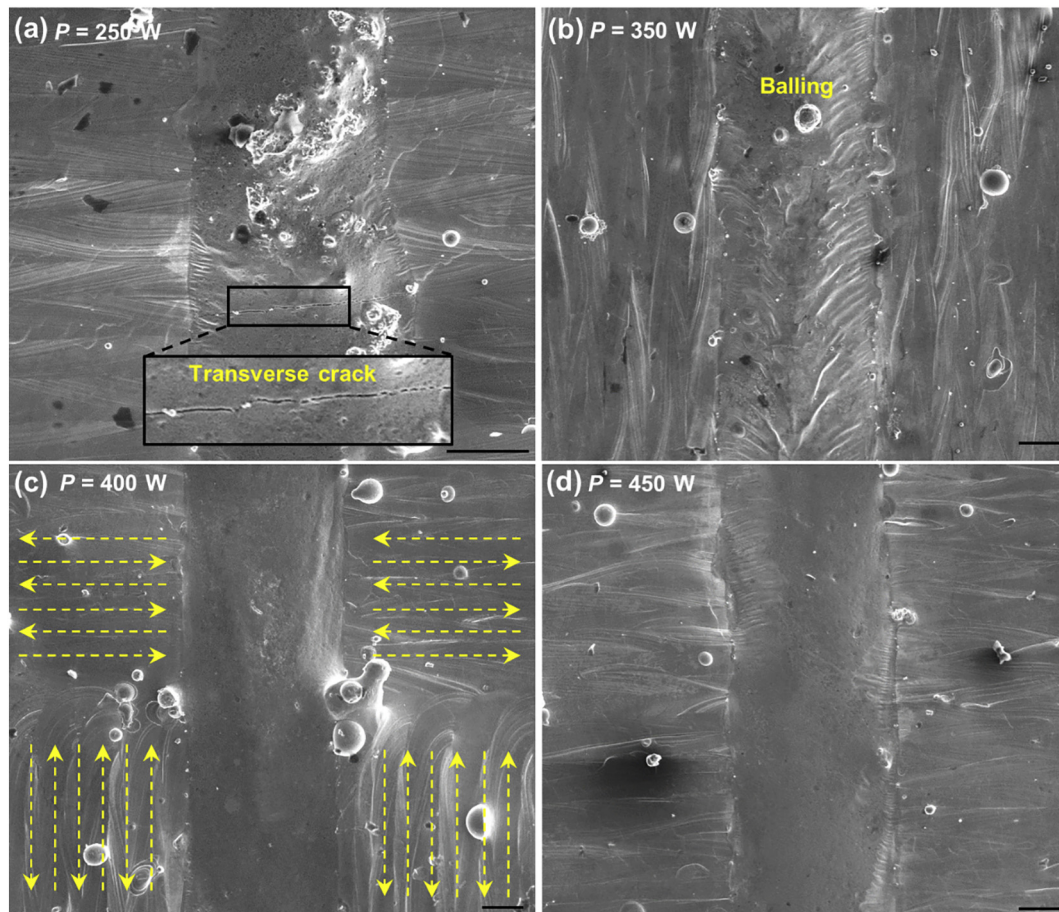


Fig. 8. SEM images showing the effects of laser power on typical surface morphology of scanned tracks under v of 300 mm/s; P of: (a) 250 W, (b) 350 W, (c) 400 W, (d) 450 W. All the scale bars are 100 μm .

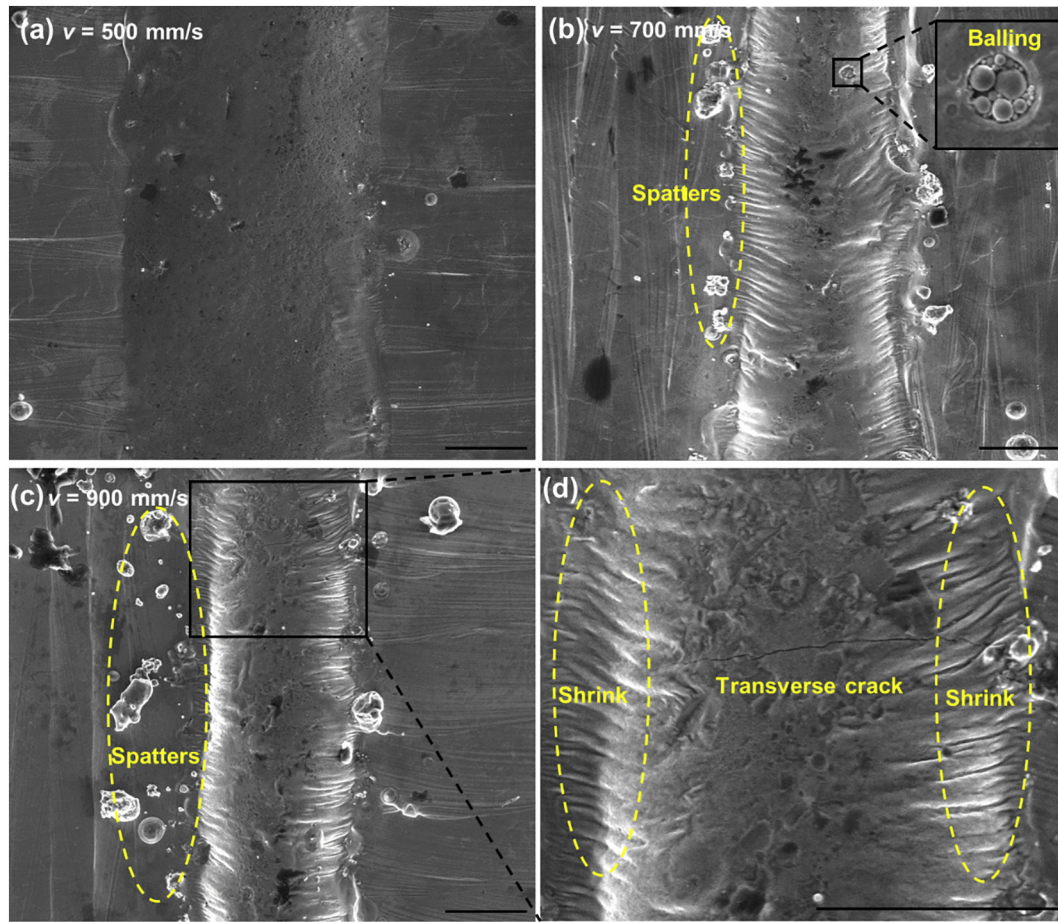


Fig. 9. SEM images showing the effects of scan speeds on surface morphology of scanned tracks under P of 450 W, v of: (a) 500 mm/s; (b) 700 mm/s; (c) and (d) 900 mm/s. All the scale bars are 100 μm .

decreased under increased LED. In this situation, the increasing wettability of scanned tracks induced the dense tracks with smooth surface. Nevertheless, when the scan speed exceeded 700 mm/s, the lesser energy absorption resulted in the incompletely melted TiB_2 particles and the balling phenomenon. This balling phenomenon can be explained by the reduced surface energy during unstable melting, which is consistent with Yadroitsev's finding that excessively high scan speed may give rise to the balling effect [34]. It can be observed that a large amount of micrometer scale balls were induced by the splash at 450 W combined with a high scan speed (inset of Fig. 9b). Meanwhile, the tracks exhibit an unstable surface consisting of excessive shrinkage and spatters at 700 mm/s or 900 mm/s, which is due to the reduction of the liquid surface energy. It is particularly noteworthy that the relatively high scan speed results in a decrease of the diameter of cylindrical tracks and even leads to a significant transverse shrink and cracks (Fig. 9d). The transverse cracks were attributed to the high residual stress and the low ductility of TiB_2 material. In general, these phenomena of balling, shrink and cracks brought about the poor surface quality, resulting in the poor interlayer bonding and the attendant interlayer pores during laser melting of $\text{TiB}_2/\text{Ti6Al4V}$ multi-material.

3.4. FEA simulations of scanned tracks

Numerical simulations of thermal behavior of the TiB_2 scanned tracks during SLM process were performed to further reveal the forming mechanisms. Fig. 10 shows the temperature distributions at different processing parameters. It was obvious that the maximum temperature at the center of the TiB_2 layer changed from

3648 $^{\circ}\text{C}$ to 3306 $^{\circ}\text{C}$ during the laser power decreased from 450 W to 250 W. This variation significantly reduced the width and depth as shown in Fig. 4b. In addition, once the scan speed further increasing, the working temperature and the diameter of TiB_2 scanned track both decreased because of the less obtained energy. The instability of the molten pool also significantly increased. Therefore, a number of droplets splashed from the surface of the TiB_2 scanned track, inducing the formation of spatters and the micrometer scale balls as shown in Fig. 9b. Fig. 11a depicts the different liquid times at different P and v through the shaded areas. The combination of high power of 450 W and low speed of 300 mm/s allowed more time (0.3 ms) for melted powder to redistribute, accordingly, achieving a continuous and stable scanned track with smooth surface as seen in Fig. 8d. For the P of 250 W and v of 300 mm/s, the shorter liquid lifetime of 0.2 ms and lower maximum temperature could result in the unmelted particles in the track. For the P of 450 W and v of 900 mm/s, a low temperature and almost no liquid lifetime increased the top surface roughness of the scanned track because of a short dwelling time of laser spot at relatively high scan speed. Moreover, Fig. 11b showed that the cooling rate elevated slightly from 27×10^6 $^{\circ}\text{C/s}$ to 31×10^6 $^{\circ}\text{C/s}$ as the laser power increased from 250 W to 450 W, and it nearly had no change as the scan speed increased from 300 mm/s to 900 mm/s. However, the relatively high scan speed yielded a rapidly changed heating and cooling rates in 1.2 ms, resulting in high thermal stresses and cracking susceptibility as shown in Fig. 9d. Therefore, a relatively high scan speed combined with the drastic cooling rate and low temperature may be detrimental to the forming stability of the TiB_2 scanned tracks on the Ti6Al4V alloy layer.

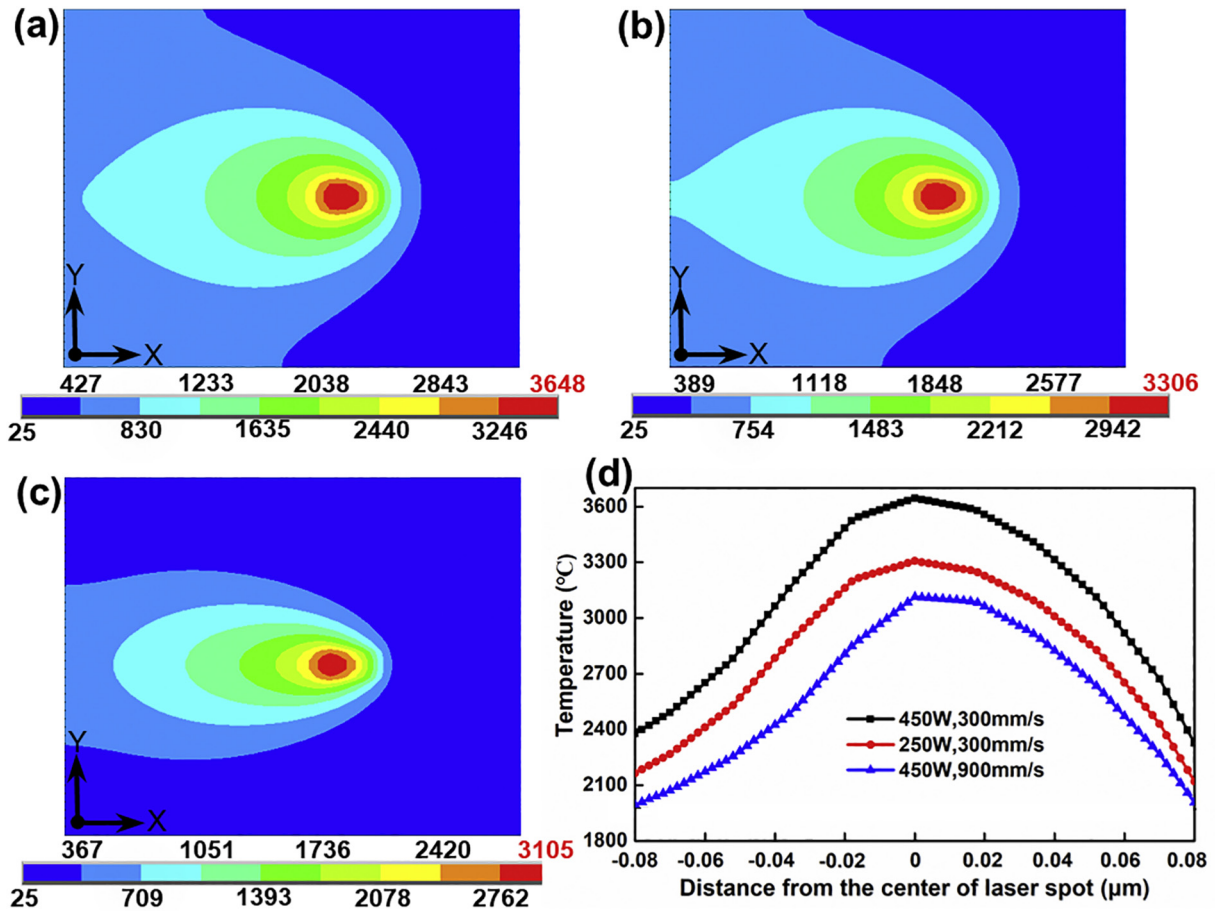


Fig. 10. Temperature distribution around the laser spot during SLM process: (a) 450 W, 300 mm/s; (b) 250 W, 300 mm/s; (c) 450 W, 900 mm/s; and (d) the temperature distribution along the x axis.

4. Conclusions

The current study investigated the processing parameters, cross-sectional features and surface morphologies of TiB_2 tracks formation on Ti6Al4V alloy during multi-material SLM. The main results can be concluded as follows:

- 1) The continuous tracks with conduction mode were easy to be fabricated in the LED range of 0.46–1.54 J/mm. Compared with the width of the cross-sections, the depth was more sensitive to the variation of laser power and scan speed. The unmelted particles and balling at the relatively low power were ascribed to the insufficient energy input and the keyhole porosity at 500 W was caused by the

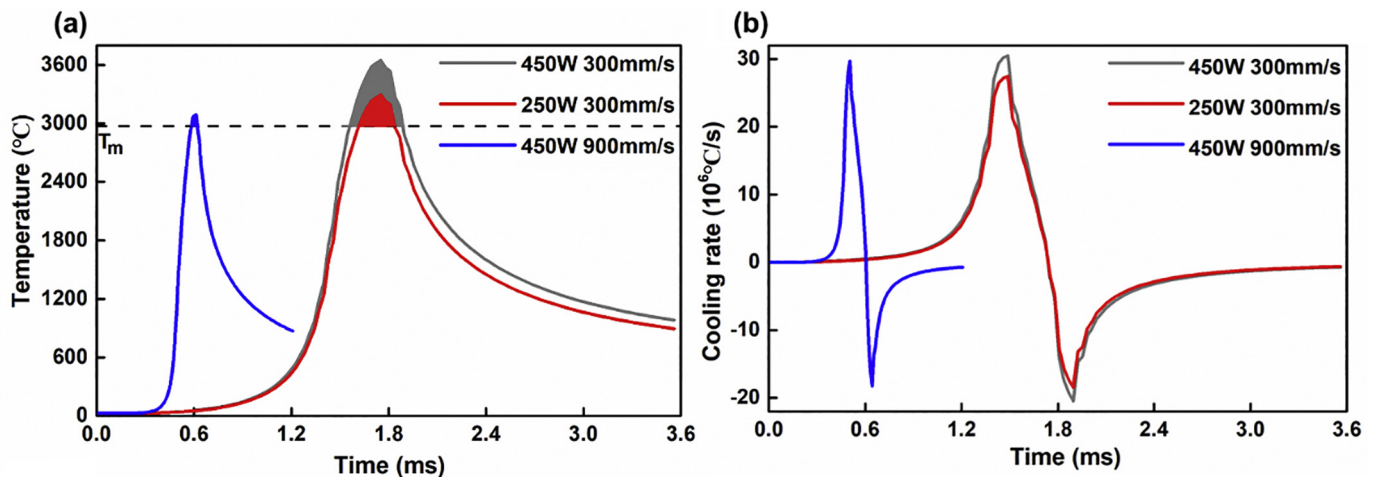


Fig. 11. (a) Temperature distribution and (b) cooling rate versus time at the center of the scan track under different linear energy densities, T_m represents the melting point of TiB_2 .

entrapped gas. The fine microstructures of the cellular grains in the center of the pool and the dendrites around the boundary were revealed.

- 2) The unmelted particles and folded surfaces were found in the surface of scanned tracks at P of 250 W–350 W. At excessively high scan speed, the tracks exhibited an unstable surface consisting of excessive shrinkage, spatters and transverse cracks. Therefore, the LED range of 0.67–1.50 J/mm ($v = 300$ –600 mm/s, $P = 400$ –450 W) was found to be a suitable scope of processing parameters to fabricate stable TiB₂ tracks with a smooth surface.
- 3) The FEA simulations also validated that the laser under high power of 450 W and low speed of 300 mm/s allowed enough melting time for TiB₂ powder to realize a continuous and stable scanned track with smooth surface. The working temperature and the diameter of TiB₂ molten pool both decreased because of the less obtained energy when the laser power was 250 W. A relatively high laser scan speed with the drastic cooling rate may result in poor forming stability of the TiB₂ scanned tracks on the Ti6Al4V alloy.

Credit author statement

Rui Wang: Experiment design, development and paper writing.

Dongdong Gu: Resources, experiment idea, paper writing, theoretical guidance and suggestions.

Caiyan Chen: Theoretical calculation, and language modification.

Donghua Dai: Theoretical guidance and suggestions.

Chenglong Ma: Experimental development.

Hongmei Zhang: Language modification and grammar correction.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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